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Fwd: v8.3.2 status update

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To: Chris Day <chris.day@wfrc.utah.gov>

Mon, Nov 3, 2025 at 2:05 PM



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From: **Chad Worthen** <cworthen@wfrc.org>
Date: Fri, Sep 3, 2021 at 12:24 PM
Subject: v8.3.2 status update
To: Jory Johner <jjohner@wfrc.org>, Julie Bjornstad <julieb@wfrc.org>
Cc: Bert Granberg <bgranberg@wfrc.org>, Bill Hereth <bhereth@wfrc.org>, Suzie Swim <sswim@wfrc.org>

Jory,

Wanted to give you an update on v8.3.2. We are in full calibration mode right now (finally) and turning through the [iterations.to](#) calibrate to the observed data. What we have updated since last we spoke:

- We recently obtained UDOT's 2019 AADT data and processed it so we can use it in our models:
 - Previously, we were calibrating to 2015 observed data and our best guess at 2019. We are now calibrating to 2019 data.
 - As part of this work, we also updated the traffic factors that convert from AADT to AWDT, seasonal variations (for AQ), and truck percents.
 - We also updated the volume forecasts coming into/out of the external gateways based on historical data that now goes through 2019 (the previous external forecasts were based on historical data that went to 2017).
 - All this should help v8.3.2 be better calibrated and more in line with v9.0 when v9.0 comes out.
- We have included all the WFRC/MAG plan amendments (highway and transit) in the model networks. We recently found an issue with the road connectivity around HAFB during calibration. This has been fixed and the fix is part of the current calibration work.
- Because v8.3.2 will be used on the Davis I-15, we wanted to check the assumptions for external trip distribution patterns. In the past, we had just assumed the behavior from previous model iterations, but they weren't updated (we'd updated the total volume coming into/out of the external gateways, but not the distribution pattern). In order to update these this round required going back to our source data and mine (or remine) the behavior out of these databases, then process it into something we could use for the model. [This one, unfortunately, had a pretty sizable impact on schedule...sorry about that.] That work is done and the external distribution pattern is being calibrated along with the rest of the model.
- Similarly we are looking at and calibrating the truck generation rates and distribution patterns. While we don't currently have the same level of data to calibrate trucks as we do for other trip types, we have some basic guide posts and traffic count data. Similar to the IX and XI external trip patterns, UDOT's (and WFRC's) interest in the I-15 EIS, we felt it best to give it some attention for v8.3.2.
- Because of these changes, we are relooking at the weights put into previous models and readjusting to fit the new data and model changes. Kind of like when you get new tires, the mechanic has to rebalance the wheels to work with the new tires, we need to rebalance the adjustment factors to work with the changed input data or process. We are right now in the process of "rebalancing" to correspond to the various changes we are making to the model.
- Here's the latest on telecommuting in the model:
 - We are pulling together the final touches on the telecommute behavior assumptions that will need to go into the model. We hope to send this out today, with follow up discussions with each agency to follow (today/next week).

- Working with all the agencies, we'll make any adjustments to the parameters to get to the final set that will be used in the base model.
- Next Thursday, Sep 9 10:30-11am, we have an IMTC meeting with the modeling tech folks and can discuss telecommute assumptions/changes. Because of the previous 1-1 agency meetings, we hopefully can come to a consensus about the assumptions that will go into the base model. I don't know if there is time to pull the policy group (IMPC) together given how hard it is to find an open spot on the IMPC group calendars. If we find one for next week, we'll meet in person. Otherwise we'll coordinate with the policy group through an email thread with the expectation/hope of arriving at a consensus by the end of next week or early the following week.
- The model will be coded (again hopefully today) based on the pattern from the External Forces work in v8.3.1. Work at Home (WAH) employment rates will be based on the pattern Julie set up earlier and adjusted to the ACS data Bert pulled together. The net WAH trips will be calibrated to the average rate assumptions agreed to (we'll start calibrating the model to the draft assumptions, that way if we have to recalibrate later on it'd be a little quicker). The WAH employment rates will be in one file and vary by year (again so we can meet the changing regional average by year). The target and the rates used in calibration constitute the "base" model that presumably would be used for multi-agency planning work. The model will also have a dial that sets a multiplier to increase/decrease the WAH employment rates, to allow the agencies to do sensitivity testing of higher/lower WAH rates. (If the user wanted to use a different WAH employment rate profile, they would switch out the rates in the input file, similar to how they would if they needed to run a specialized SE scenario.)
- The other external forces (CAV & eCommerce) will be in the model and will use the rates and behavior from what came out of the group meeting recommendations, unless we hear otherwise. We can see if there is time at the upcoming IMTC/IMPC meetings to discuss CAV/eCommerce, but there's a good chance that the telecommute discussion will take most/all of the time. Because CAV & eCommerce affect future behavior, not 2019, these wouldn't have an impact on our current calibration work, so we have a little more time if adjustments/changes need to be made to the rates/implementation strategy.
- As we go through the calibration process, we will pull together comparisons to past model versions and note where there is significant change, and will share this with WFRM LRP/Analytics and our partners. If we are all good with the new model, then we will finish the documentation and release the model. Because this coordination may take a little time, projects needing to use the model (like I-15 or RTP) could begin to use the "final draft" v8.3.2 model. It should be pretty stable at this point, but if any further edits need to be made to the model based on partner feedback, we'll work with the pre-release users when the final, updated model is ready and go over what changed and if/how the changes would affect their project.
- It's a short week next week, but I think we will be done (or close to) with most of the calibration by the end of the week, with the exception of the comparisons to earlier versions and maybe discussion around CAV/eCommerce.

I know we are not moving as fast as we had hoped or promised. Thanks for your patience! We're moving as fast as we can while still trying to maintain a sure footing for our modeling products (that sounds a little like a sales pitch, but I think the idea or sentiment is right on). We're really sensitive to your schedule, so please keep communicating your stress level regarding using the new model so we can adapt what we are doing accordingly (as best we can).

Chad



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